

Four clinical ECG sources

PTB-XL | Georgia | Chapman | Ningbo

66,379 retained records, 13 harmonised labels, 10 s at 100 Hz

leave-one-source-out: train on three, test on the fourth, four rotations

10 encoder families x 2 size budgets (+1 ablation)

compact ≤ 32 KB int8 weights | standard ≤ 128 KB

each family scaled to the largest width that fits,

everything else pinned: loss, optimiser, epochs, seed

88 trained models

RQ1 does the family decide accuracy?

external macro-AUPRC and AUROC per held-out source,
float32 and int8, record-level bootstrap intervals

int8 graphs, QDQ, calibrated on training sources

RQ2 does the efficiency claim hold?

ST Edge AI $\rightarrow$ arm-none-eabi $\rightarrow$ flash over SWD

measured Flash, peak SRAM and DWT latency; 22 of 22 ran

measured against predicted

RQ3 could we have known in advance?

graph-derived activation proxy against the SRAM
the allocator actually required